

Technical Architecture

Secure, evidence-aware digital twin architecture for auditable energy analysis and investment decisions.

Team EnerGen AI · Project Irene
T1 — AI for Clean Energy



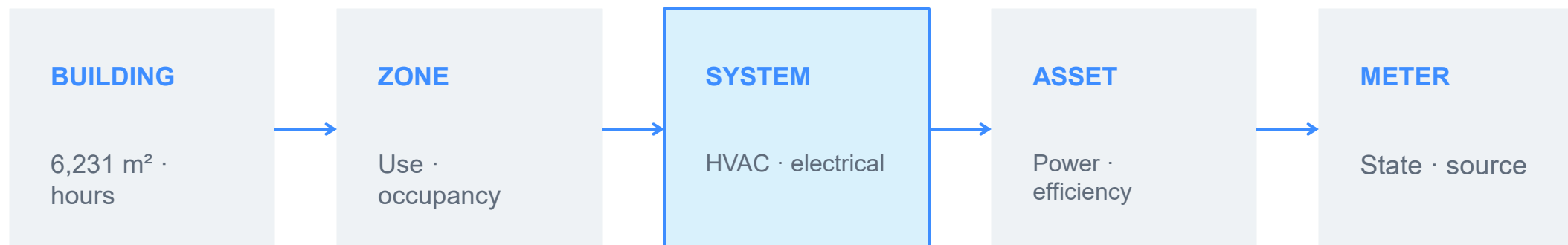
Irene connects through governed adapters



Migration: inventory → map → backfill → reconcile → shadow run → approved cutover. API/SQL/SFTP/gateway connectors are configured per client.

Evidence links every asset and value

Entities link physical scope, operating state, engineering constraints and the evidence behind every value.



The anonymized reference case links approved aggregates, synthetic public records and quality evidence to a building-system-meter graph and 106.14 kWp of PV.

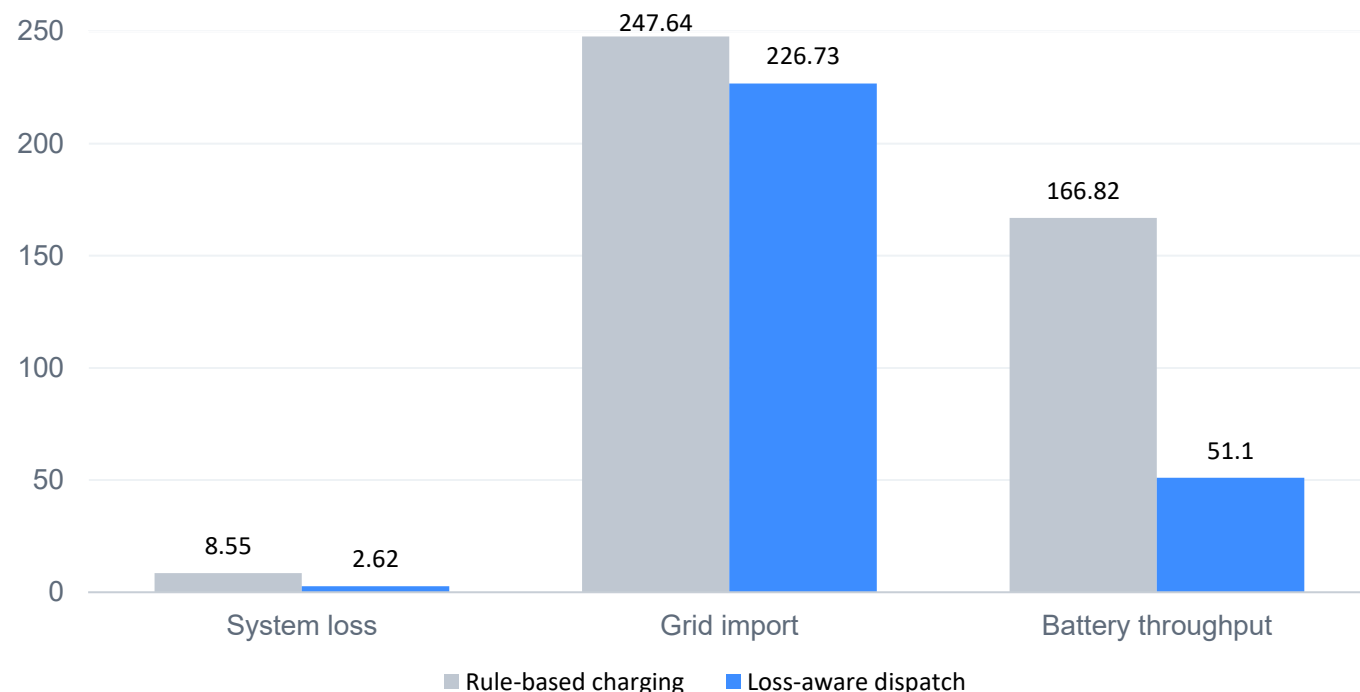
Evidence quality governs every model layer

MODEL LAYER	AVAILABLE EVIDENCE	DATA REQUIREMENT	VALIDATION CHECK
Energy baseline	12 approved monthly anchors	15-min meter + schedules	Month sums + residuals
Forecasting	Calendar prior · 14.36% NMAE	Interval load + indoor sensors	Time split · MAE · RMSE
Digital twin	Reference graph + confirmed client project	Zone/circuit/equipment map	Source trace + mapping + constraints
Optimisation	Storage dispatch optimisation	BESS design + supplier quote	Cost · loss · SOC · violations

Evidence boundary: 14.36% NMAE validates a synthetic reference time-shape only; site-hourly load remains estimated until authorized interval data are available.

Loss-aware optimisation reduces storage losses

Design study only: the reference case has no installed battery. Both strategies use identical constraints and terminal state of charge.



69.4%

lower conversion loss

69.4%

less battery throughput

8.4%

less grid import

One governed stack supports audit, pilot and portfolio scale



Target customers: commercial buildings · campuses · industrial parks · energy service companies (ESCOs). Local-first by default; client files stay governed.

Measured first. Optimised second.

Validated on a Ningbo reference case; designed for configurable deployment in Malaysia, pending local pilot validation.

IMPLEMENTATION PATH

File audit → temporary metering → calibration → savings verification → multi-site scale.